A ZFC Verification of the Two-Dimensional Case of Swift's $P(\omega_1)$

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Source and Target

The source paper is Andrew Swift, *Coarse embeddings into* $c_0(\Gamma)$, arXiv:1703.01891. In Section 3 Swift defines $P(\alpha)$: for every $n \in \mathbb{N}$ and every coloring

$$\sigma : [\alpha]^n \to \mathcal{C},$$

at least one of the following alternatives holds:

1. there is a sequence $(F_j)_{j=1}^{\infty}$ of pairwise disjoint members of $[\alpha]^n$ such that $\sigma(F_i) = \sigma(F_j)$ for all i, j;
2. there is $F \in [\alpha]^{n-1}$ such that
$$\sigma\big(\{F \cup \{\gamma\} : \gamma \in \alpha \setminus F\}\big)$$
is infinite.

Problem 2 in Section 5 asks whether Theorem B is true for ω_1, i.e. whether $P(\omega_1)$ holds.

Combining (12) and (13) we finally obtain that for some $D \geq 1$, and any $k \in \mathbb{N}$,

$$\frac{1}{D} \max_j |a_i| \leq \| \sum_{i=1}^{k} a_i e_{\gamma_i} \| \leq D \max_j |a_j|. \tag{14}$$

for all $\{\gamma_1, \ldots, \gamma_k\} \subset [1, \Lambda)$, which proves our claim. $\square$

5 Final comments and open problems

Let us mention in this final section some problems of interest.

First of all, we do not know whether or not Theorem A is true if we replace λ by smaller cardinal numbers.

Problem 1. *Assume that X is a Banach space with* $\mathrm{dens}(X) \geq \omega_1$*, and assume that X coarsely embeds into $c_0(\Gamma)$ for some cardinal number Γ. Does X have trivial co-type? If moreover X has a symmetric basis, must it be isomorphic to $c_0(\omega_1)$?*

Of course Problem 1 would have a positive answer if the following is true.

Problem 2. *Is Theorem B true for ω_1?*

Connected to Problems 1 and 2 is the following

Result

This packet proves the first nontrivial arity case. It is a partial result only: the full question requires all finite n.

Theorem 1. *The $n = 2$ instance of $P(\omega_1)$ holds in ZFC. Equivalently, for every set of colors $\mathcal{C}$ and every map*

$$\sigma : [\omega_1]^2 \to \mathcal{C},$$

either there are pairwise disjoint two-element sets $F_1, F_2, \ldots \in [\omega_1]^2$ with $\sigma(F_i)$ constant, or there is an ordinal $\alpha < \omega_1$ such that

$$\{\sigma(\{\alpha, \beta\}) : \beta < \omega_1,\ \beta \neq \alpha\}$$

is infinite.

Proof Intuition

The obstruction to the second alternative says that every vertex of the complete graph on ω_1 sees only finitely many colors. Attach to each vertex the finite set of colors incident to it. Since every edge has a color visible at both of its endpoints, these finite sets are pairwise intersecting. The Delta-system lemma turns an uncountable family of pairwise-intersecting finite sets into an uncountable family with a fixed nonempty finite root. On the corresponding uncountable set of vertices, every edge color lies in that one finite root. The ordinary finite-color Ramsey theorem on a countably infinite subset then gives an infinite monochromatic clique, and hence an infinite monochromatic matching.

Proof

Assume that the second alternative fails. Then for each $\alpha < \omega_1$ the incident color set

$$C_\alpha = \{\sigma(\{\alpha, \beta\}) : \beta < \omega_1,\ \beta \neq \alpha\}$$

is finite.

For distinct $\alpha, \beta < \omega_1$, the edge color $\sigma(\{\alpha, \beta\})$ belongs to both C_α and C_β. Hence

$$C_\alpha \cap C_\beta \neq \varnothing$$

for all distinct α, β.

Apply the Delta-system lemma to the uncountable family $(C_\alpha)_{\alpha<\omega_1}$ of finite sets. There are an uncountable set $A \subset \omega_1$ and a finite root R such that

$$C_\alpha \cap C_\beta = R$$

whenever $\alpha, \beta \in A$ are distinct. Since the family is pairwise intersecting, $R \neq \varnothing$. Moreover, for all distinct $\alpha, \beta \in A$,

$$\sigma(\{\alpha, \beta\}) \in C_\alpha \cap C_\beta = R.$$

Thus the complete graph on A is colored using only the finite color set R.

Choose a countably infinite subset $B \subset A$. By the infinite Ramsey theorem for finite colorings of pairs, there are an infinite subset $B_0 \subset B$ and a color $r \in R$ such that

$$\sigma(\{\alpha, \beta\}) = r$$

for all distinct $\alpha, \beta \in B_0$. Enumerate $B_0 = \{\gamma_0, \gamma_1, \gamma_2, \ldots\}$, and set

$$F_j = \{\gamma_{2j}, \gamma_{2j+1}\} \qquad (j = 0, 1, 2, \ldots).$$

The F_j's are pairwise disjoint two-element subsets of ω_1, and all have color r. This is the first alternative of $P(\omega_1)$ for $n = 2$. $\square$

Verification Notes

The proof uses only two standard ZFC combinatorial facts: the Delta-system lemma for an uncountable family of finite sets and the ordinary infinite Ramsey theorem for finite colorings of pairs on a countable set. The conclusion matches exactly the $n = 2$ specialization of Swift's property $P(\alpha)$.

The argument does not prove the $n \geq 3$ cases. For $n = 3$, for example, failure of the second alternative gives finite color sets on pairs, while a triple color must be compatible with all three of its pair-faces. The simple vertex-root Delta-system reduction above no longer directly produces a finite global color set on an uncountable vertex set.

Novelty and Search Bounds

This packet should be treated as a modest partial result, possibly folklore. The run indexes were searched for arXiv id 1703.01891, the title phrase, and core phrases around $P(\omega_1)$, Theorem B, and coarse embeddings into $c_0(\Gamma)$; no existing run packet for this target was found. Local corpus search found Swift's source paper and a separate use of $P(\omega_1)$ as a hypothesis, but no local packet or later source resolving Swift's full Problem 2. A newer local nonlinear-geometry book notes Swift's embeddability theorem and related Q^{ω_1} material, but did not supply a full answer to this problem.

Human-Review Recommendation

Review as a likely valid but narrow partial result. The main review question is not the proof itself, which is short, but whether this $n = 2$ case is too elementary or already explicitly recorded in the literature to count as more than duplicate memory.

References

[1] A. Swift, *Coarse embeddings into* $c_0(\Gamma)$, arXiv:1703.01891.

[2] J. E. Baumgartner, *Canonical partition relations*, J. Symbolic Logic **40** (1975), 541–554.